

Seismological Data Analysis

Linux

- Multitasking
 - Multiple access (Parallel Computation)
 - Unix is bug free, stable, licensed
- Characteristic

Kernel

→ Core of operating system

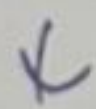
→ Memory or time allocation

Interpret Commands

Shell

Command Interpreter

Command



Shell



Kernel

Display

- csh shell
- bash shell
- tcsh shell

} Almost Similar

Utility

Commands we are typing on the computer.

→ ~ → home directory

\$ → End of command prompt

→ pwd → print working directory (or folder)
 ↓
 address where we are.
 → in linux

→ ls → listings → contents of the directory

→ cd → change directory

cd directory name
command space argument

→ cd ~ → go to home directory.

→ cd ../ → go one directory back
 ↘ 1 previous directory

→ cd ../../ → two directory back.

→ man → manual common

→ man ls → display manual for command.

→ ls -a → also show some hidden files
 name start with "."

→ ls -l → column wise display of files
 & folders.

d in starting → it is a directory

d - - - on file for current
 permissions people.

r → read

x → execute

w → write

another → group

→ xr - * read but not write, but can execute.

→ u g o u = user, g = group &o = others

→ ls -la → permissions for hidden files.

→ cd . → Keep in same directory.

→ mkdir director1 → make new directory with name director1

→ rm dir director1 → remove director1

→ seismograms, we can type seis and hit tab it will take the name.

→ mv → ^{rename} move command (Similar to cut)

→ mv a.txt z.txt change a.txt to z.txt

→ cp copy command

→ cp z.txt a.txt copy z.txt to a.txt but keep z.txt

→ Create a file echo "I love" > sda.txt
create a new file

→ cat sda.txt show content inside this file

→ head from top how many lines we can see

→ head -2 file name → display top 2 lines of the 'file name'

→ tail -2 file name → display bottom 2 lines of the file name

→ clear → clear the command

→ ls geo* → & display only those files whose name starts with geo

* → ~~character~~ wild

Character

*

Function

wild wildcard for any no. of characters in a filename

?

wildcard for single character in a filename.

\$ reference = variable

& ~~if~~ a = 0 then \$a = value of a

& executes a command in background

; Separates commands on the same line

> redirection of standard output

→ \$ * can follow anything

geo * txt dat

only 'geo' matters nothing after *
doesn't matter.

→ ls geo?.txt

whatever there is for
? will be displayed.

→ \$ matlab & open matlab & run in background but still we will be able to use the terminal.

→ \$ shutdown -h now → shutdown the system

→ edit files 'gedit'.

→ \$ vi → editor

→ \$ vi . name of file

→ less → display one screen at a time

→ tar → tape archive

→ echo → make file

→ echo "hi" > new_SDA.txt
new file

→ ls -l new_SDA.txt → tell when the file was formed.

→ gzip new_SDA.txt → zipping the file
extension .gz

→ gunzip → unzip

→ cp new_SDA.txt new_SDA-copy.txt → two new files will be created.

→ tar cvf → create a file (cvf)

→ tar cvf new_SDA.tar SDA-TRASH
Create a tar file named → for this file

→ tar xvf SDA-TRASH → untar this file.

→ top → processes currently running, with
PID → Process identification

→ kill PID → stop process PID

→ kill -9 PID → forcefully kill stop process PID

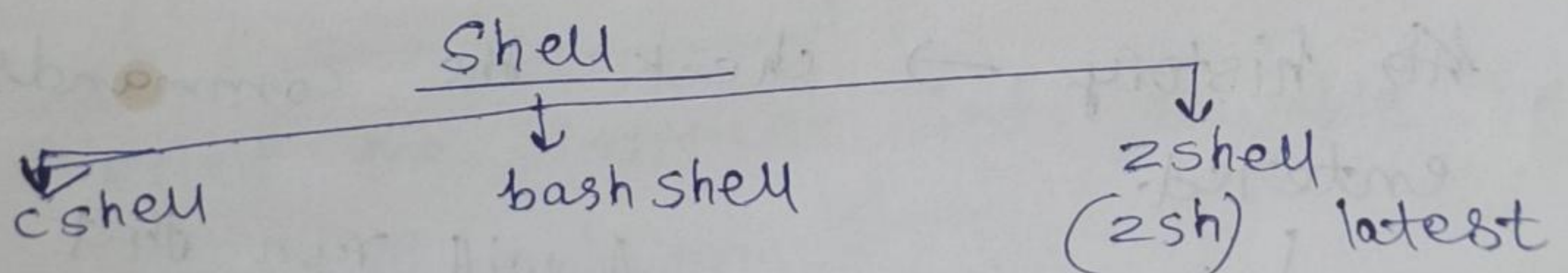
→ ps -u name → process only i am
using (user → "name")

→ whoami → currently logged in user of
the system.

→ vi filename → open a file

→ grep filename search filename → search for
word "search" in "filename" and display
all the line names that do contain them.

→ rmdir → remove a directory.



→ echo \$SHELL → wild card for variable SHELL

→ cd → go to home directory.

cd ~
cd ~username

→ .bashrc → whenever we are opening any terminal it will automatically run.

→ man ls → give information (manual) for ls, what this ls can do.

→ gnome-open → opening a pdf

→ vi editor  command mode escape
Insert mode 'I'

→ o → open a line below

O → open a line above

→ Escape → : → q quit without saving

→ Escape → : → wq quit with saving.

→ ~~Alt~~ history → check the commands previously entered.

↳ !Sr No → ! will run that command.

→ !! → run last command.

AWK

→ special programming language for seismological data acquisition.

→ Alfred Aho Peter Weinberger Brian Kernighan

→ AWK reads the entire catalogue in the row wise pattern.

→ Columns → Fields , Rows → Records

→ `awk -F "1" '{ print $2 }'` → print 2nd column.
defining a field separator

→ `awk -F "1" '{ print $2 }' filename > t.t`
redirect this file to a new file

→ `vi t.t` → open this file.

→ `print '{ print "lat=" $3 }'` filename.txt

print the latitude inf. available in 3rd column.

AWK Variable Name

What it stands for

FILENAME

name of current
file

RS

Input Record Separator
(Default is new line).

OFS

Output field separator
string (Blank is
default)

ORS

Output Record Separator
string (Default is new
line)

NF

Number of fields in
input record

NR

Number of input record

OFMT

Output format of number

FS

Field Separator character
(Blank & Tab is default).

awk -F "1" '{print \$2}' filename.txt.

→ `awk -F "1" NR > 1 { print $2 }`

only 2nd column not 1st column.

→ `awk -F "1" ' $11 > 3.5 { print "lat=" $3, "long=" $4, "mag=" $11 }'`
11th column is magnitude
file name

only 3.5 magnitude lines will be displayed.

→ `awk -F "1" ' $6 ~ /ISC/ { print "lat=" $3, "long=" $4, "mag=" $11 }'` file name

String comparison, earthquake author only
ISC.

→ `awk -F "1" ' { print $3, $4, $11 }' file`

> new file name

Print the 3, 4, 11th column and will print 1st row also in new file rather than showing on screen.

→ waveform data → .sac seismic analysis code.

→ `ls *.sac` → list all files with .sac Extension

→ `ls *.sac > t.t` → list and direct to file

→ `'a, 'bs|.sac //g`
↓ first line ma
last line mb
with .sac extension to global nothing.

→ `awk -F "1" "BEGIN {x=1} NR>1 {`
print "Earthquake - "x++,"\$3,"\$4,"\$5,"\$11"}"

delimiter
begin a variable
filename.txt
Earthquake - 21 Column 3 Column 4 Column 11
Column 11

→ `awk -F "1" '{if NR>1, print $3,$4,'}`
file name
Column 3 Column 4
Space

→ `vi a.txt` say `q`
→ `vi b.txt`
b1
b2
b3

→ to merge → vi paste a.txt b.txt

⇒
a1 b1
b1 b2
c1 b3

→ Say x1.txt vp = 1 x2.txt vp = 4
vs = 2 vs = 5
p = 3 p = 6

→ join x1.txt x2.txt > x3.txt
cat x3.txt

⇒
vp 1 4
vs 2 5
p 3 6

→ cut -d - -f2 y1.txt
telling delimited "-" and printing only numerical value.

Station component network component
GMU UU HHZ

→ echo "I love Seismology" > sanjay.txt
make file sanjay.txt with content
I love Seismology.

→ echo "I hate playing" >> sanjay.txt

It will append this line after "I love seismology". → This will be in 2nd line.

→ after any command we can redirect > file name

⇒ How to access Earthquake data from management system

→ Collect → Clean → Archive data

↓
Disseminate to users

→ Strong motion data (collected from Strong motion seismometers) used for seismic hazard analysis. It gives acceleration data, to study near field effect from an earthquake. Not very sensitive to ground vibration.

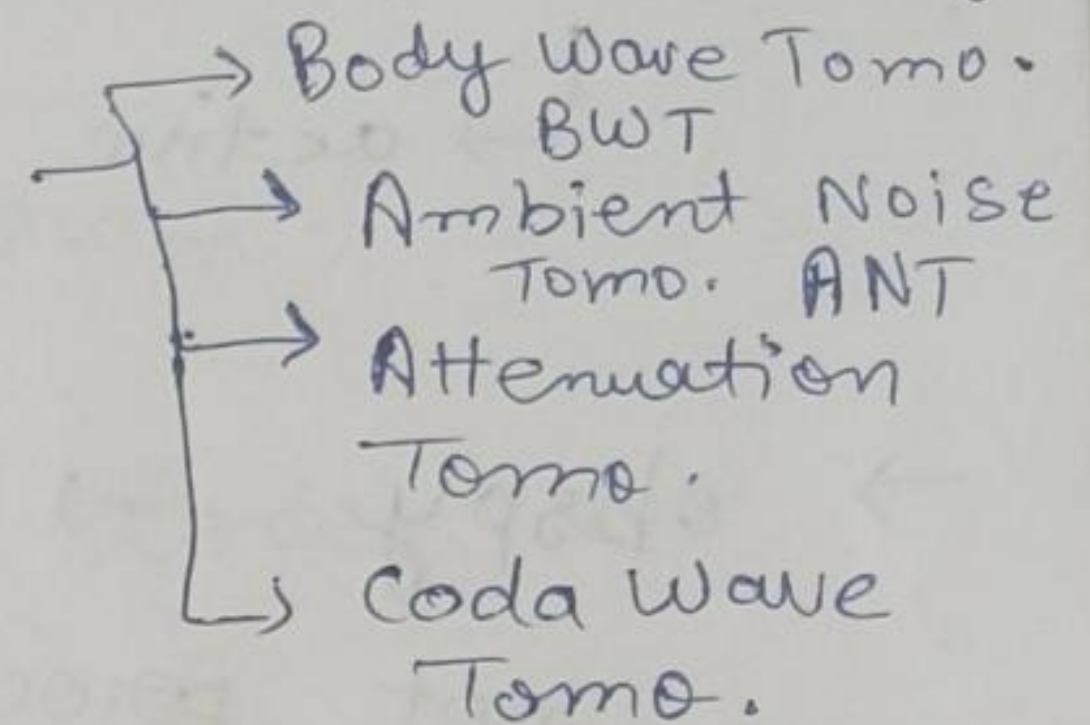
→ Broadband data: Record near & far data. They generally clip at large amplitudes. It will not record true value of earthquake due to clipping of higher amplitudes.

Why do we need data?

1.) We can do near field study using Strong motion data.

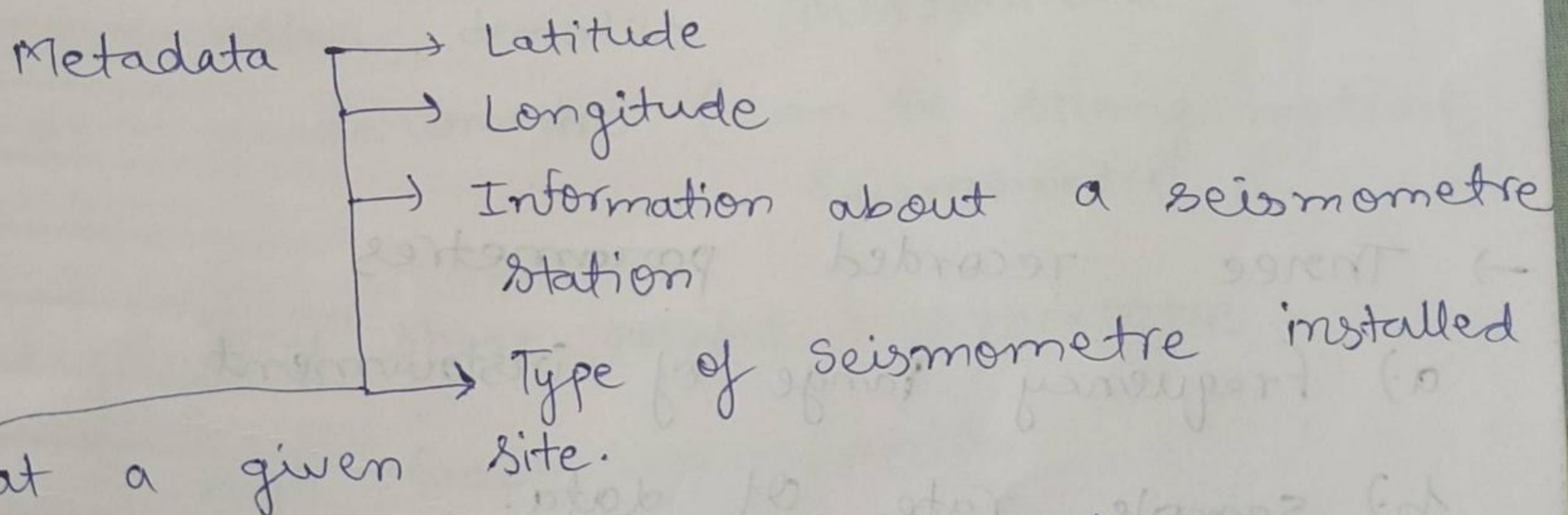
2.) * For studying movement analysis, focal mechanism.

3.) Particular kind of phase. For eg SKS
(S wave in mantle $\rightarrow$ K phase in outer core $\rightarrow$ S wave in mantle $\rightarrow$ recorded at station) Splitting,
Receiver function, Tomography



4.) Continuous records for noise study.

waveform $\rightarrow$ Time Series



$\rightarrow$ Length of time series data.

$\rightarrow$ Sample Rate, Channel (Vertical or Horizontal)

$\rightarrow$ Instrument Response to Ground motion.

Data formats

IRIS

archive data

$\rightarrow$ SEED format

Standard for Exchange of Earthquake Data.

→ mini SEED or mSEED

rdSEED → Command useful to read SEED format data

→ SEG-Y Society of Exploration Geophysicist
↳ active source seismology.

→ obspy → Python toolbox for reading data, not processing.

↓ Convert to a processing friendly format

Linux based ← SAC (Seismic Analysis Code)

Software Both active & passive seismic

→ Three recorded parameters

a) Frequency range of instrument

b) Sample rate of data.

c) Channel.

Network Code
Station Name Code
Channel

→ 2014 1532222. BN. DAK. BH2
Year Julian Day Hours Minutes Broadband High Gain 2 component

→ Network Code : Area where the instrument is installed has a particular ins code.

→ EHE, EHN, ELE, ELN, ELZ, BHE, BHN, BNZ, HNE, HNN, HNE, HNN, HNZ, ~~B~~ DPE, DPN, DPZ,

Last extension can be these.

1st letter represents band code

H → High broadband

D → Very Very short period

E → Short period.

2nd letter denotes instrument code

N → accelerometre (can be strong motion seismometre)

P → Very short period Seismometre

H → High Gain Seismometre

L → Low Gain Seismometre

→ Broadband Seismometres are generally high Gain seismometres.

3rd Letter indicates orientation

Z → Vertical

N → North South

E → East west

→ Data Request Tool

SOD → standing order of data

Arms

→ find_events → query for what eq data is available.

→ find_channels → for what channel, data is available (information about stations).

→ find_seismograms

1st we find ~~channels~~ ^{events}

→ find_channels -R -115/-109 / 42/46 -c BHZ

↓
Seismograms

Specify region

Longitude -115 to -109

Latitude

42 to 46

specify channel
BHZ

→ find_events -b 2000 -m 8.0

Beginning year

magnitude

↑
redirecting output to channels

→ find_seisvents -b 2000 -m 8.0 -r | find_channels

-R -115/-109 / 42/46 -c BHZ -r | find-seisogram

Beginning time

end

-b -1 ttp -E .2 ttp -c BHZ

→ sac

SAC > r filename
↓
read

SAC > p1

p2 all seismograms
will be plotted at
top of each other.

p → plotted one after
another.

→ $x = \text{"Seismology"}$

echo "I love x " → I love Seismology

echo 'I love x ' → I love x

" " → used to execute commands in the
bash Script

Today = 'date', echo "Today's date is" \$Today
today's date is date

Bash Variables

vi

• bashrc

→ Hidden bash file.

1.) Environmental Variables

cd \$ Bash Environment

2.) User defined Variable

String Variables

Numeric Variables

Array of String Variables

• $x = 1$, $y = 2$ echo $\$x + \y

class = "Seismology"
myfavouriteclass = \$class

→ 1 + 2 (It will not do maths, String).

• echo myfavouriteclass
↓
myfavourite class

• awk is utilized for Column Operation.

• Adding Integers

$b = \$((1+2))$ → 3

a is defined as string variable then

$a = b + 2$

$a = \$((b+2))$ → 4

Similar to

$z = \$((a+b))$

let $z = a + b$
↳ command

$b = \$((1+2)) = \text{let } b = 1+2$

$a = \$((b+2)) = \text{let } a = b+2$

Adding Decimals

$c = \$ a + 1.13$ Not understood

$c = \$ a + 1.13 | bc$ ✓

! bin / bash

$a = 2$

$b = \$((a+3))$

$c = \$((4+3))$

$\text{let } d = b+4$

$e = 'echo \$d + 3.14 | bc'$

$\text{echo } \$a \ \$b \ \$c \ \$d \ \$e$

→ 2 5 7 ~~5.14~~ 9 12.14

if no $|bc$ is used then

→ treated as a string

2 5 7 9 9 + 3.14

to see output $\$ bash filename.sh$

Array of String Variables

No space in between = sign.

days = (mon tues wed thurs fri)

0 1 2 3 4

→ In bash array counting start from 0.

echo \${days} → calling an element
→ mon

echo \${days[3]} → thurs

echo \${day[0]} → mon

→ Bash is case sensitive.

no = 10

No = 11

nO = 12

echo \$no \$No \$nO

→ 10 11 12

→ For calling array only {} are used.

year = 2020 , month = 12 , day = 30

output = \$ \$ year - month - day → 30
echo \$output

It treats year as one string

output = \${year} \${month} \${day}

echo \$output

→ 20201230

Command Line : Argument

! bin / bash

→ This question will be asked

echo -n "How many records in this file do you want to skip"

read nlines

→ input we give will be read as variable nlines

echo \$nlines

Control Structures

Are means of changing the flow of the shell script under certain conditions.

For eg.

if [condition A] then

[execute a series of instructions]

else if [condition B]

[execute a different series of instructions]

else [if A & B are not exist]

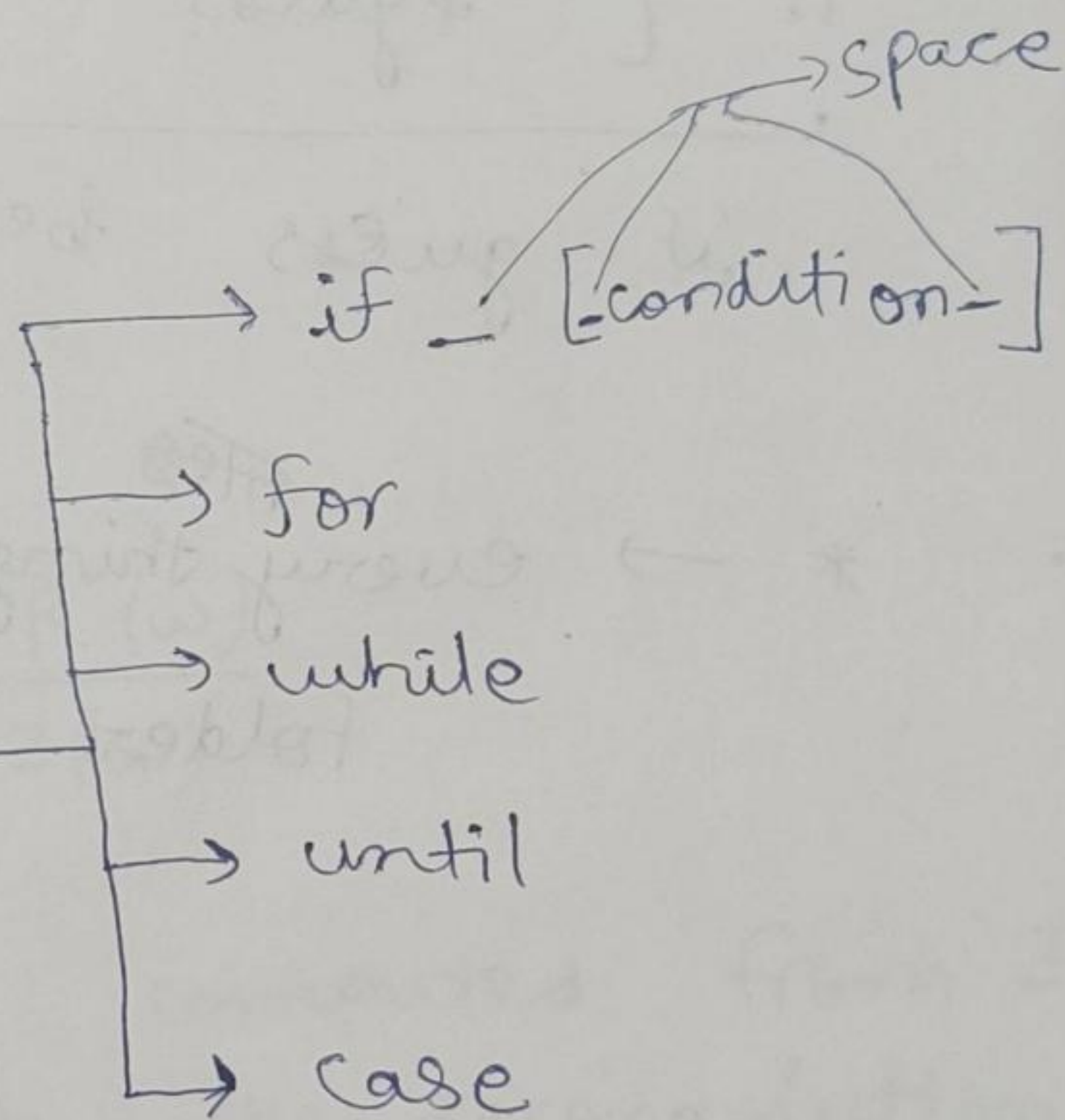
[do something else]

end

There are other control structures too

```
while [condition exists]
[do something]
else
or
```

```
for [condition]
[do something]
end
```



• Condition structures

• To compare Strings use =
Numbers use -eq

• `$guess -lt 6` → if is less than (`-lt 6`)

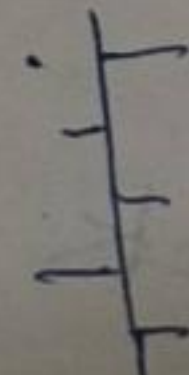
• `-e` → exist, used to check if file exist.

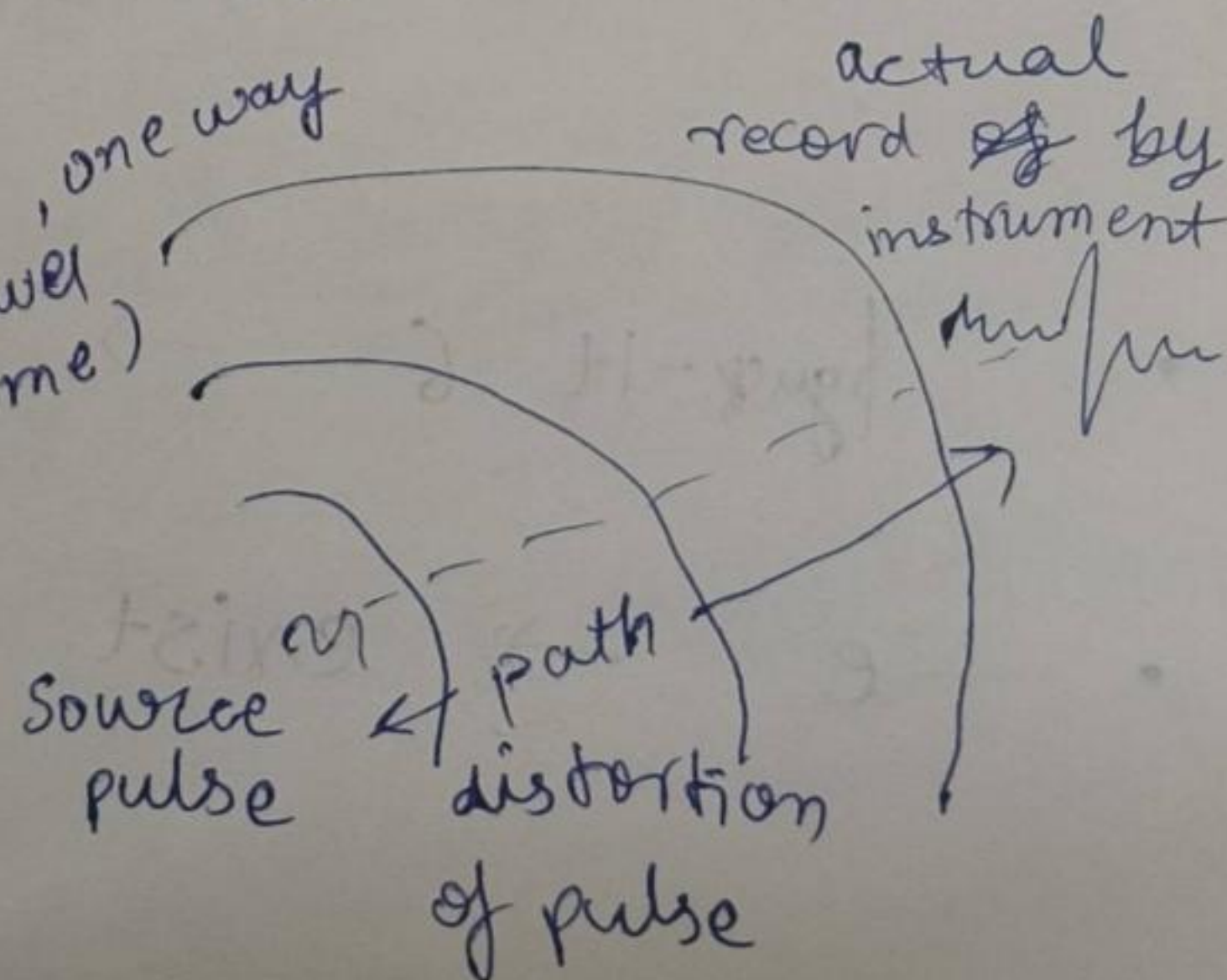
• `-d` → to check if directory exists.

- until $\rightarrow$ until that condition ~~beca~~ is true
- while $\rightarrow$ until that condition becomes true.
- $\text{if } [\text{"\$guess"} -gt "6"]$

 if guess becomes equal to 6.
- * $\rightarrow$ every thing ^{files (files)} in the current file folder

Receiver function

- To extract properties of subsurface.
- Computed from teleseismic earthquake (2500-15000 km), $>30^\circ$ away from station.
- Body waves ^{Plane wave approximation (plane wavefront)} are generally used.
- Time series of earth.
 (Ref. seismology $\rightarrow$ active source, TWT, here passive source, one way travel time)
 S * path * instrument
 = seismogram
 Interested
 Reflectivity Series = 



- Vertical Component $\rightarrow$ Signature of Source
- Radial Component $\rightarrow$ Significance of Source
+ Reflectivity Series
- Deconvolve both $\rightarrow$ we get reflectivity Series
In frequency domain

$Z \rightarrow$ source signature

$R \rightarrow$ source reflectivity

$$\text{Receiver function} = \frac{R(\omega)}{Z(\omega)}$$

- Time Series which are computed from 3 component seismogram after the deconvolution of vertical component from radial component.

Types $\begin{cases} \rightarrow \text{Radial Receiver function} \\ \rightarrow \text{Transverse Receiver function} \end{cases}$

- Radial Receiver function = $\frac{R}{Z}$

- Transverse Receiver function = $\frac{T}{Z}$

Radial Receiver function

→ Used to find shear reflectivity velocity (mostly) of the Earth.

→ Does not give relative velocity.

→ Most sensitive towards the impedance contrast.

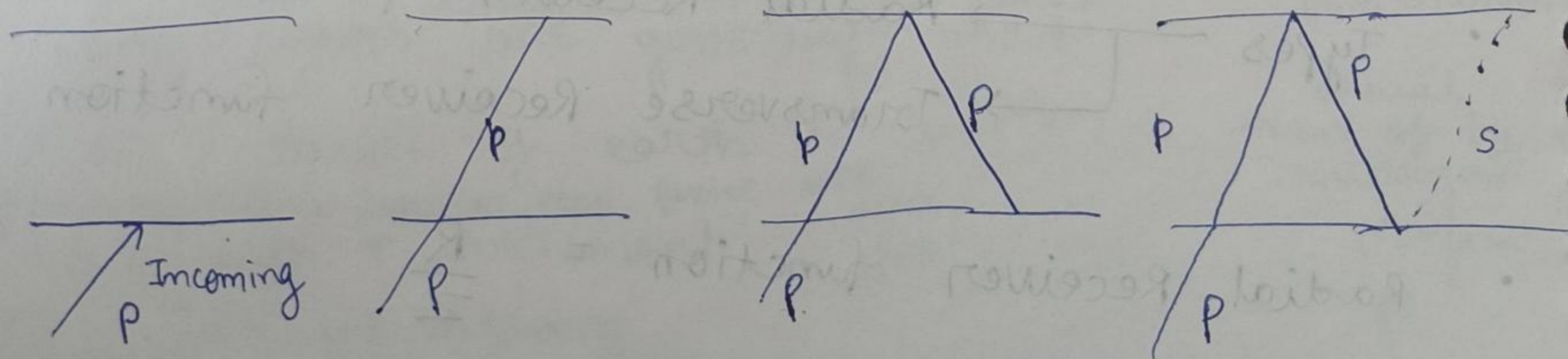
→ Depth of boundary

Transverse Receiver function

→ Used to find Anisotropy.

• Incident energy is denoted as Capital letter (P).

• Upgoing → Smaller letter, Downgoing → Capital letter.



P p l s — mode
↓
downgoing converted

- With average velocity model, depth to the discontinuity can be determined.

- P wave Coda $\rightarrow$ P wave arriving then its multiple is coming, together they are known as P wave Coda.

- Ray Parametre $p = \frac{r \sin i}{v}$ (for spherical earth),

Along the ray, it remains Constant.

r = radial distance b/w layers, for shallower earth, r is almost equal.

$\rightarrow$ Vertical Receiver Functions : S to P converted phases are ~~from~~ find out.

$\rightarrow$ LAB : Lithosphere Asthenosphere Boundary

$\rightarrow$ Swave : Larger wavelength, more sensitive to deeper part, used to find deeper parts of earth.

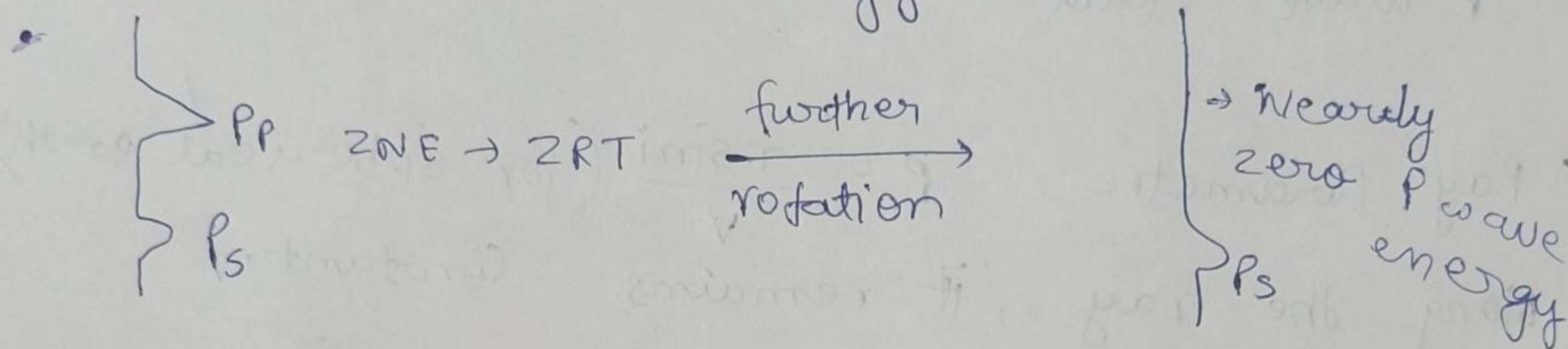
$\rightarrow$ P wave : Crustal study, small wavelength, Less sensitive to deeper part.

$\rightarrow$ Seismogram = Impulse * Source response of instrument * function * Impulse response of local earth

- $S = \frac{Z}{R}$

- Incidence Angle 'i' for S & P waves are calculated as using :

→ Direct P wave energy should be zero



$i = 0 : 0.1 : 30^\circ$
 ↓
 Generally

Take 0 as i

then calculate L, Q, T

$\Downarrow$
 $\frac{R}{Q} = \text{Receiver function}$

Similarly we do for all $i = 0.1, 0.2, \dots, 30^\circ$

then we check at what 'i' value 'p' wave energy is minimum, then we chose

L, Q, T → then deconvolve to get finalised Receiver function.

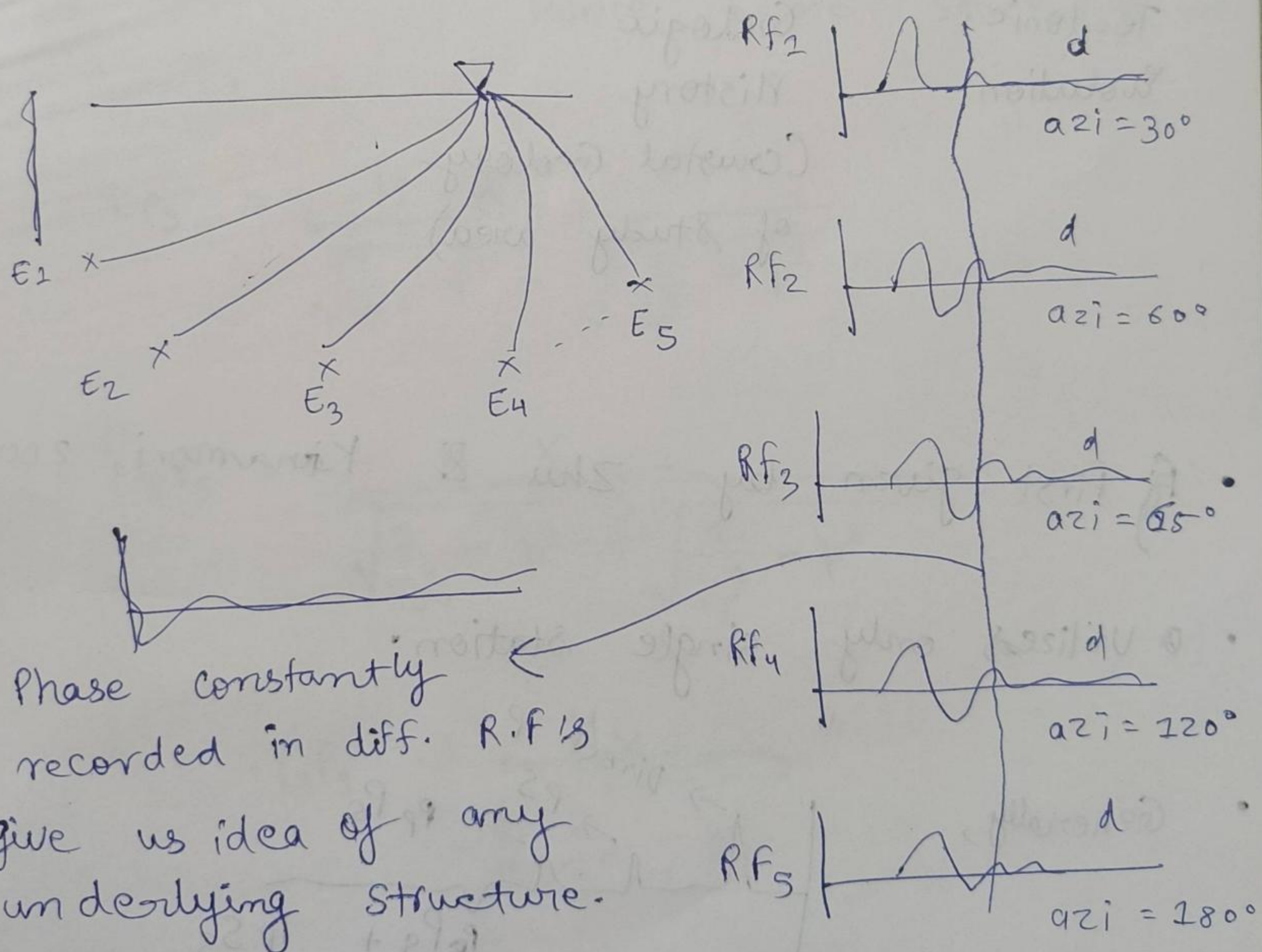
- Z → Source, R → Source + receiver path

- 1.) Assumed R.f * Z Comp. = R comp. (R_0)
 $E_0 = R_0 \sim R$ Compare with actual Radial Component
- 2.) Modified assumed receiver function = R_1
 $E_1 = R_1 \sim R$ Compare with actual

$E_0 > E_1 \Rightarrow$ keep updated R_1

$E_0 < E_1 \Rightarrow$ do not update R_0

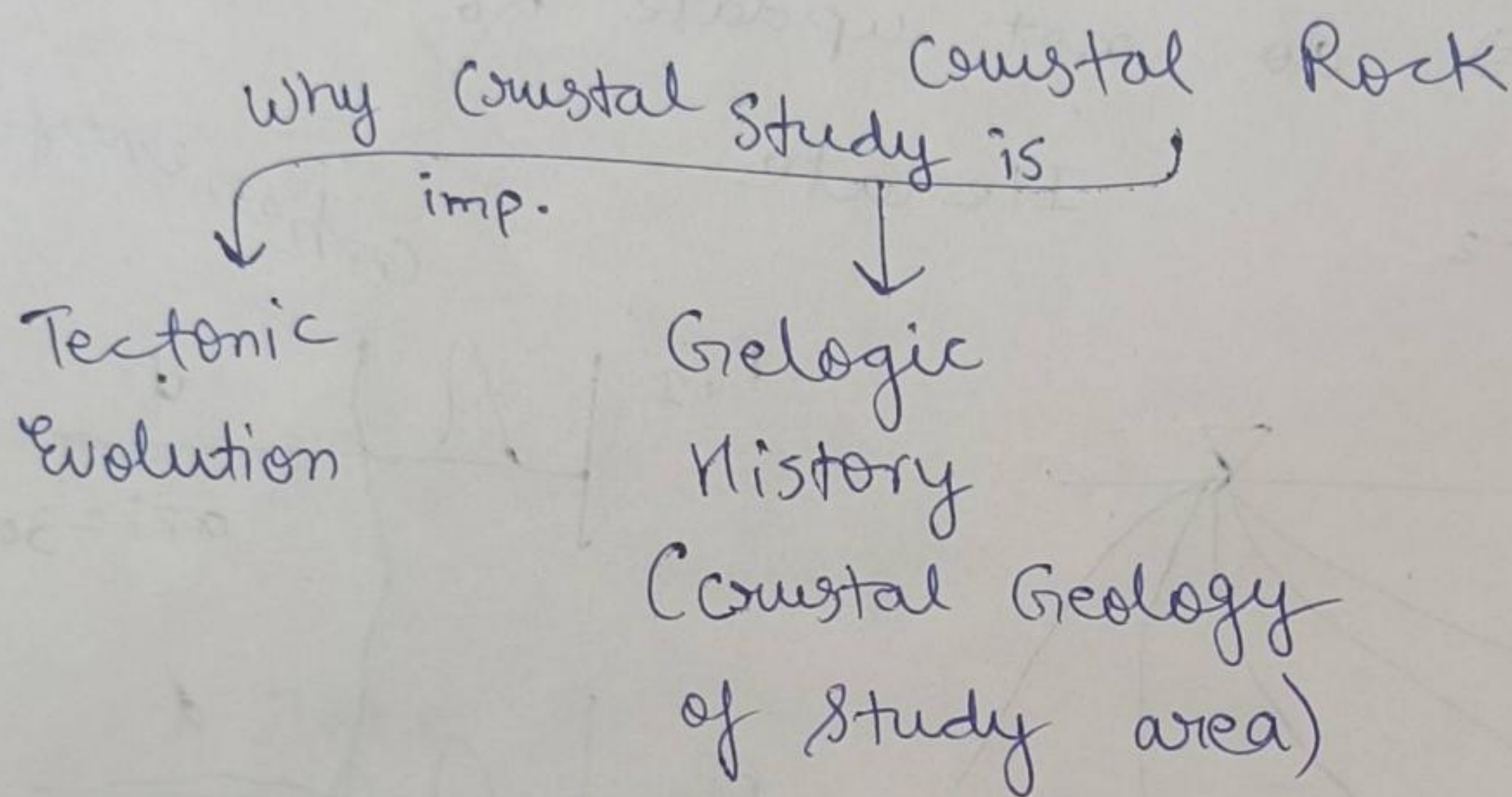
$E_1 > E_2$ Iterate



H - K Stacking

- $H \rightarrow$ Crustal Thickness

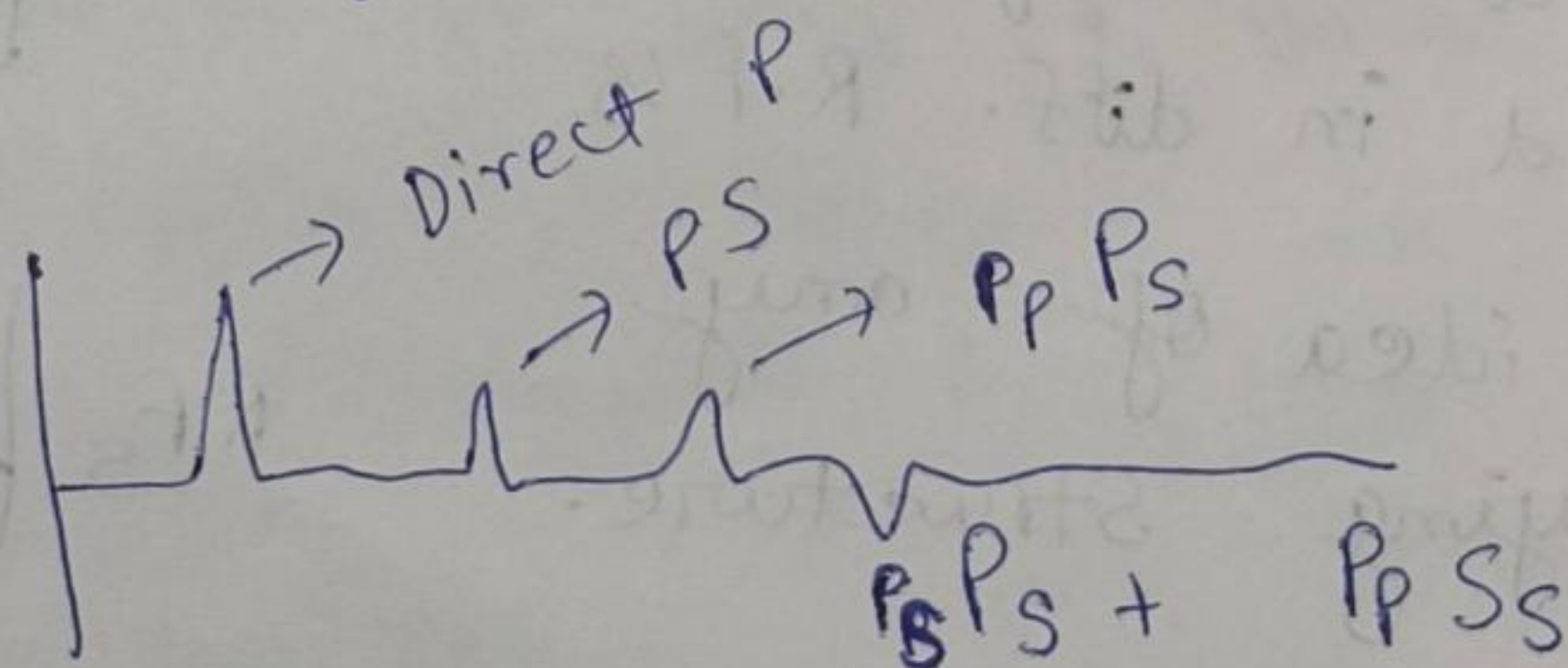
- $K \rightarrow V_p/V_s$ ratio or Bulk (average V_p/V_s)
for entire crust not a layer or point
wise but gross. $\downarrow$ Sensitive to the
Chemical
Rock Composition of the



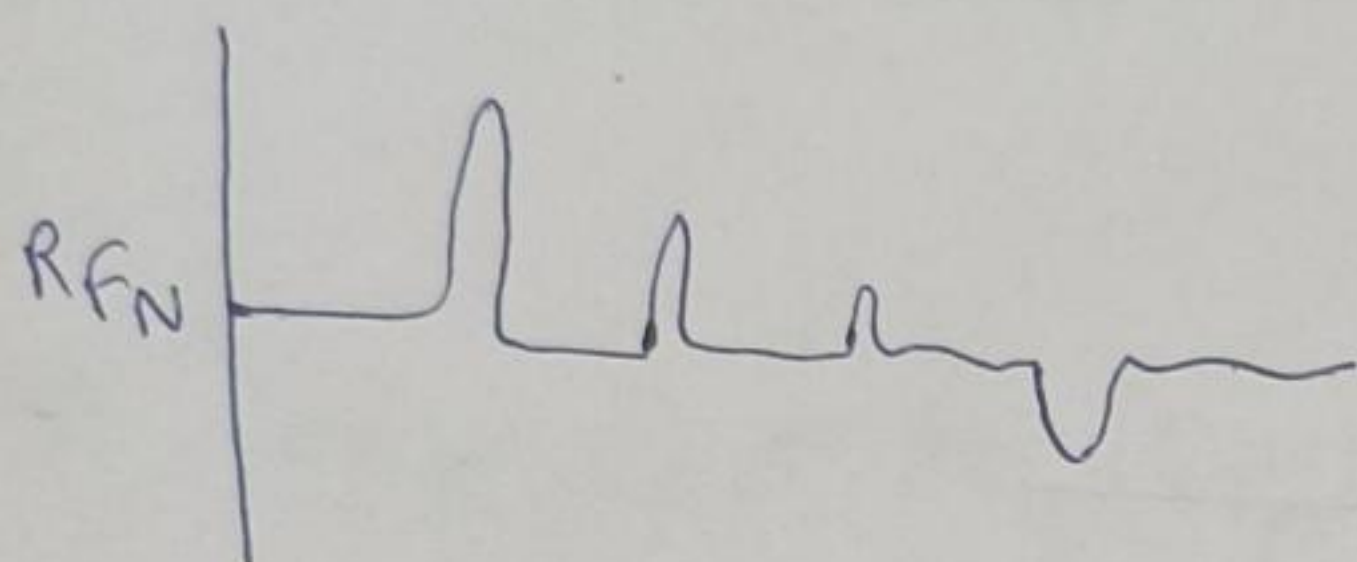
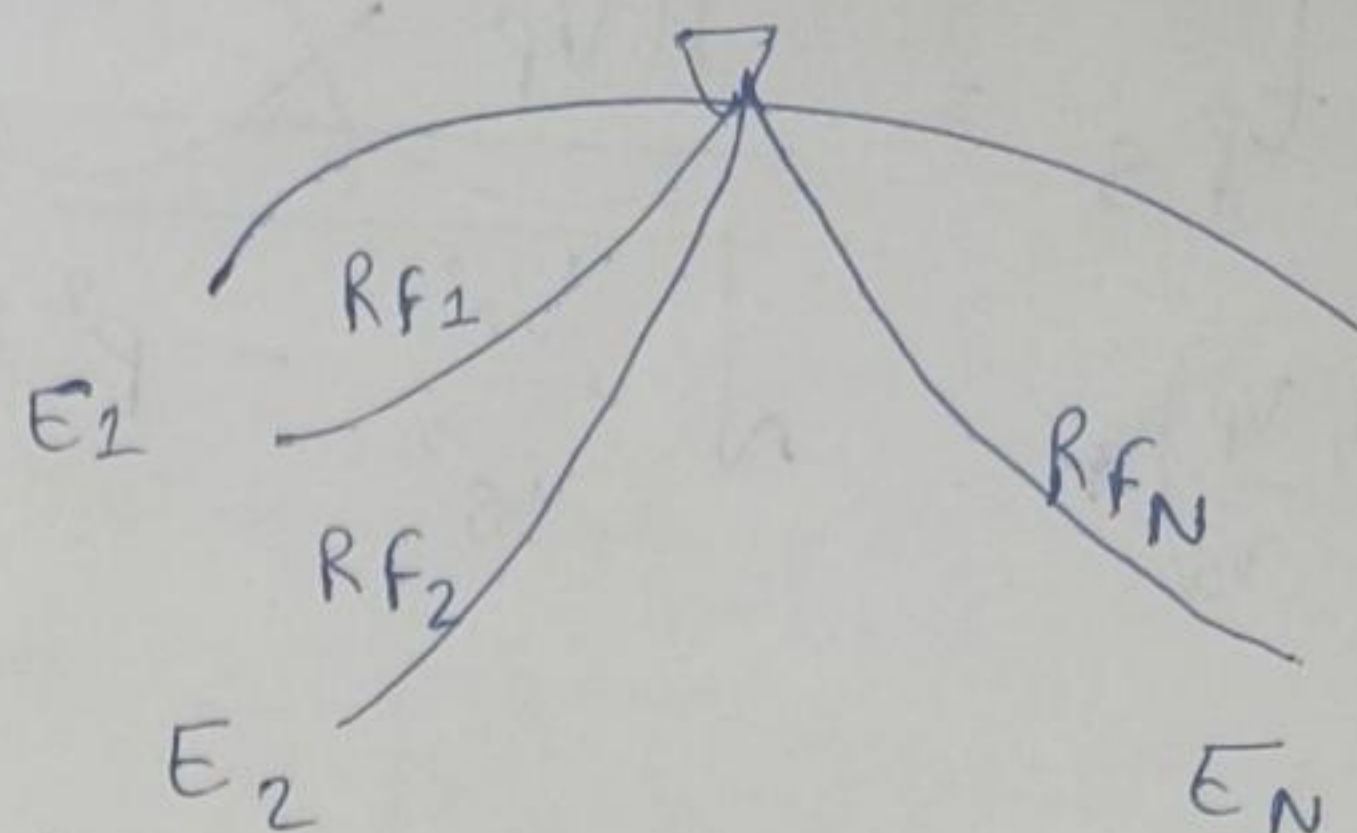
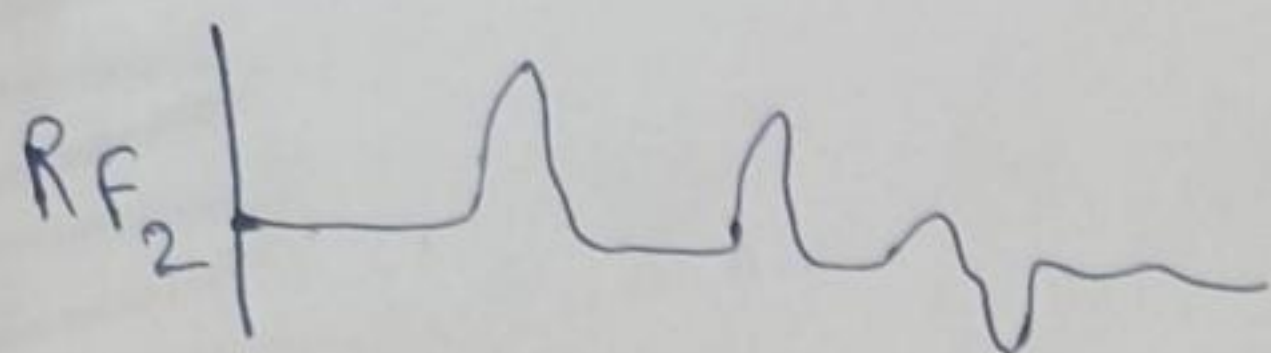
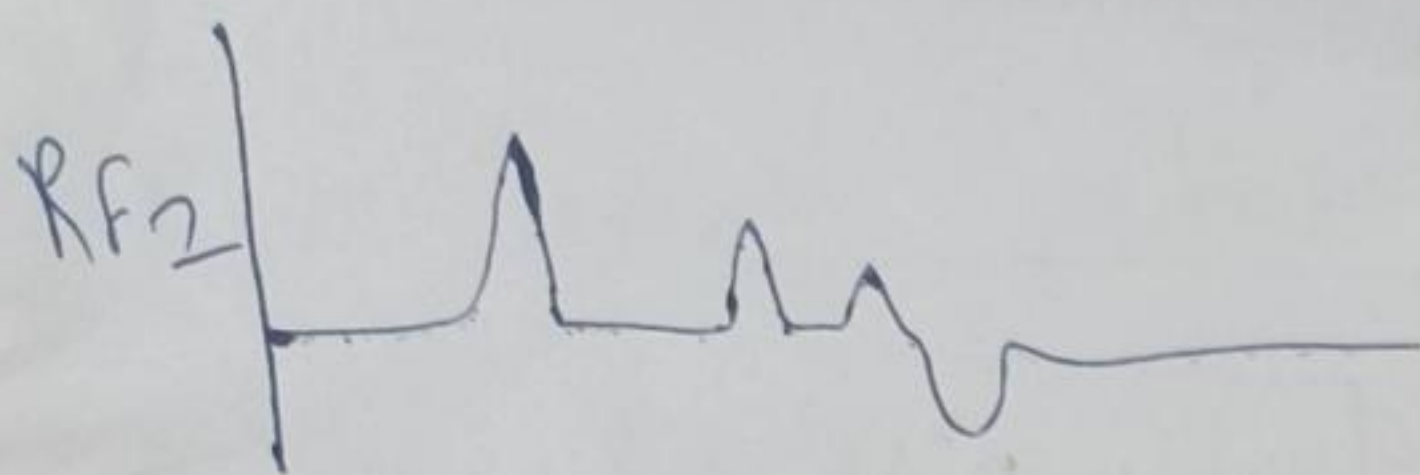
- F_0 first given by Zhu & Kanamori, 2000.

- Utilizes only single station.

- Generally,



1-) For each earthquakes, we compute the receiver function.



There will be moveout
due to the time delay
b/w the signal arriving.
Hence before stacking we

Correct it.

$$t_{ps} = \frac{H}{\sqrt{\frac{1}{v_s^2} - p^2} - \sqrt{\frac{1}{v_p^2} - p^2}}$$

$$t_{p_s p_s} = \frac{H}{\sqrt{\frac{1}{v_s^2} - p^2} + \sqrt{\frac{1}{v_p^2} - p^2}}$$

$$t_{p_s s_s} + p_s p_s = \frac{H}{2\sqrt{\frac{1}{v_s^2} - p^2}}$$

v_p = P wave velocity } Unknown

v_s = Shear wave velocity }

H = Crystal Thickness

p = Ray Parametre (known for Receiver function)

$$t_{ps} = \frac{H V_p}{\sqrt{\frac{V_p^2}{V_s^2} - p^2 V_p^2} - \sqrt{1 - p^2 V_p^2}}$$

$\left(H, \frac{V_p}{V_s}\right)$

$$t_{pP_s} = \frac{H V_p}{\sqrt{\frac{V_p^2}{V_s^2} - p^2 V_p^2} - \sqrt{1 - p^2 V_p^2}}$$

$\left(H, \frac{V_p}{V_s}\right)$

$$t_{pP_s} + P_s P_s = \frac{H V_p}{2 \sqrt{\frac{V_p^2}{V_s^2} - p^2 V_p^2}}$$

→ we try to keep $V_p \rightarrow 6.0$ to $6.6 \frac{\text{km}}{\text{s}}$
based on geology.

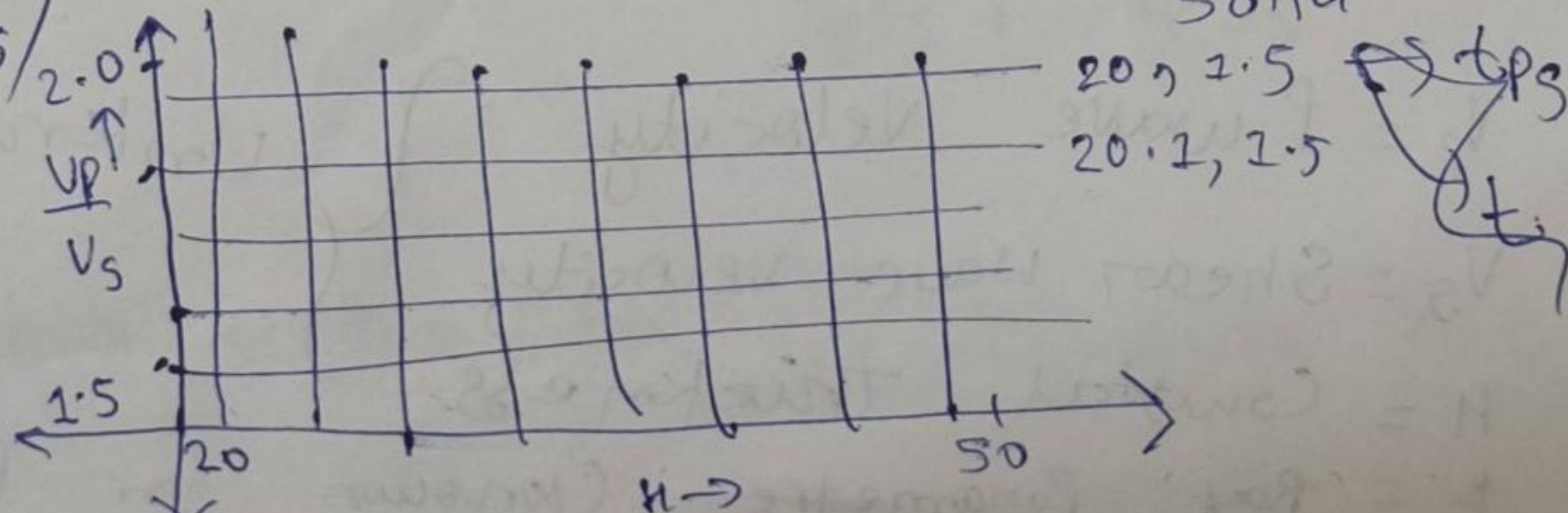
→ we are trying to make a grid.

$H = 20$ to 50 km (Moho depth)

$H = 20:0.1:50$ (20 to 50 with spacing of 0.1)

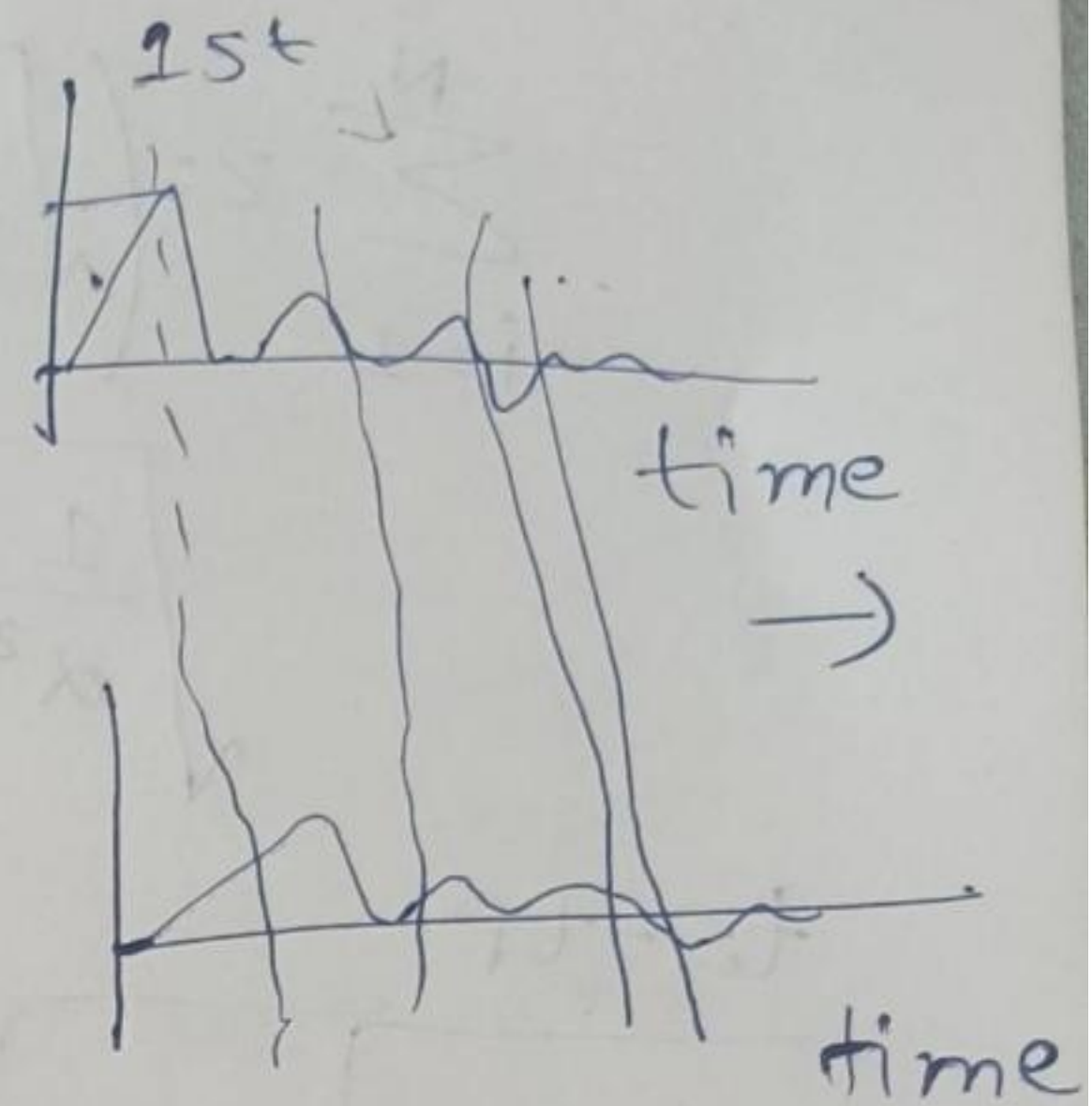
$K \left(\frac{V_p}{V_s}\right) \rightarrow 1.5:0.2:2.0$

Poisson $\frac{V_p}{V_s} = 1.73$
Solid



→ For each point we find out value of t_{ps} , t_{ppls} , $t_{ppps} + p_{pls}$.

So say for 1st Receiver



→ for each phase there will be some amplitude

$$S(H, K) = A_{ps}(t_{ps}) +$$

$$\rightarrow S(H, K) = W_{ps} A_{ps}(t_{ps}) + W_{ppps} A_{ppps}(t_{ppps} + p_{pls}) + W_{pls} A_{pls}(t_{ppls} + p_{pls})$$

at particular time, t_{ps} , t_{ppls} , $t_{ppps} + p_{pls}$ we put amplitude & weights (hit & trail).

Generally $W_{ps} = 0.5$, $W_{ppps} = 0.3$, $W_{pls} = 0.2$ acc. to geology of area. For each pair of H, K we calculate $S(H, K)$, where this value becomes final answer. $W_1 \geq W_2 \geq W_3$
 $W_1 + W_2 + W_3 = 1$

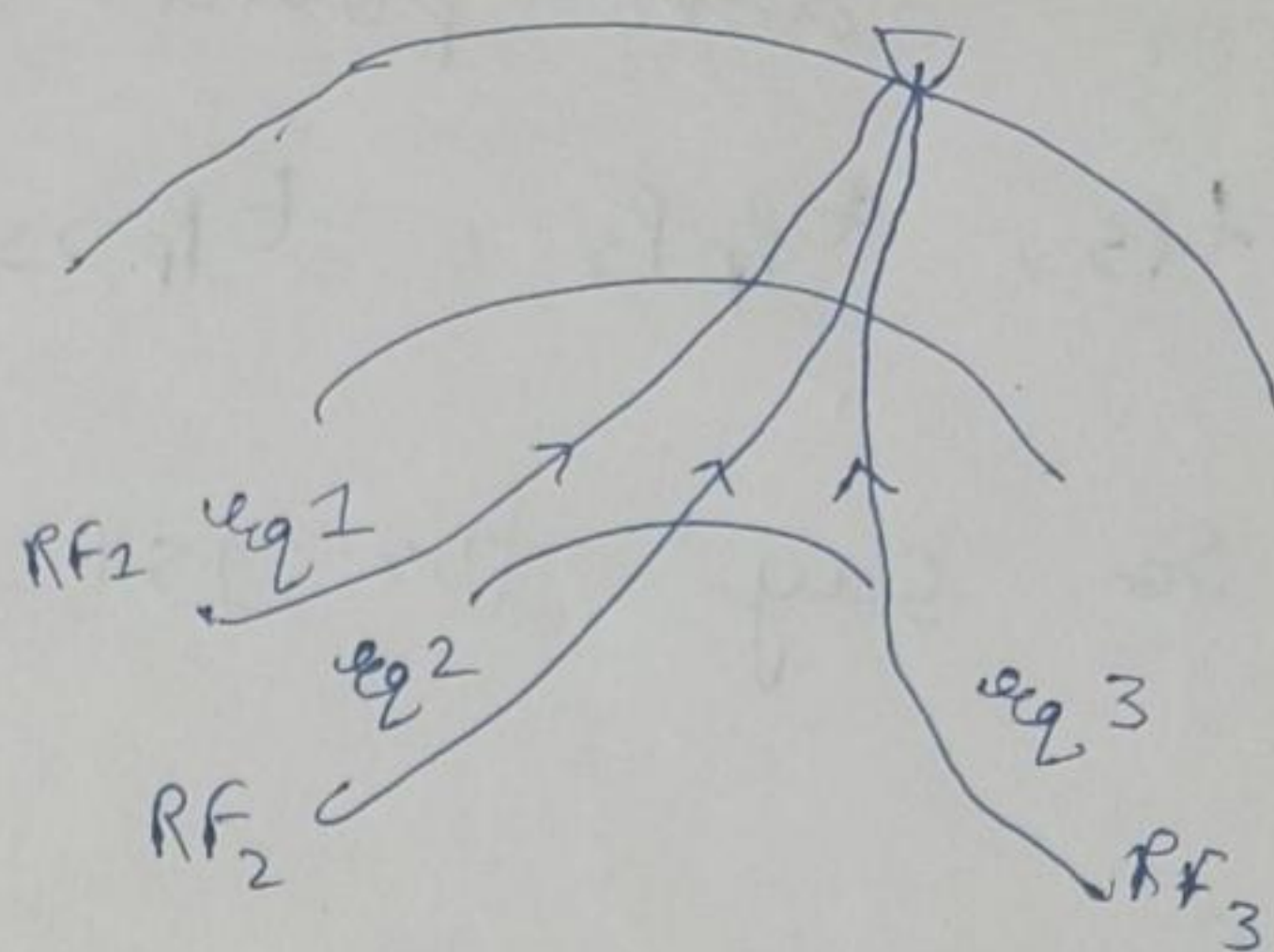
Velocity Analysis of R.F.

w. of direct arrival we are having the converted phases time.

RF = Ps receiver function

$$t_{ps} = t_s - t_p$$

$$= \sum_{i=1}^N Z_i \left(\sqrt{\frac{1}{\beta^2} - p^2} - \sqrt{\frac{1}{\alpha^2} - p^2} \right)$$



$t_s - t_p$

0.16	0.16
0.2	0.36
0.3	0.66

we place the direct arrival here & stretch for the other and

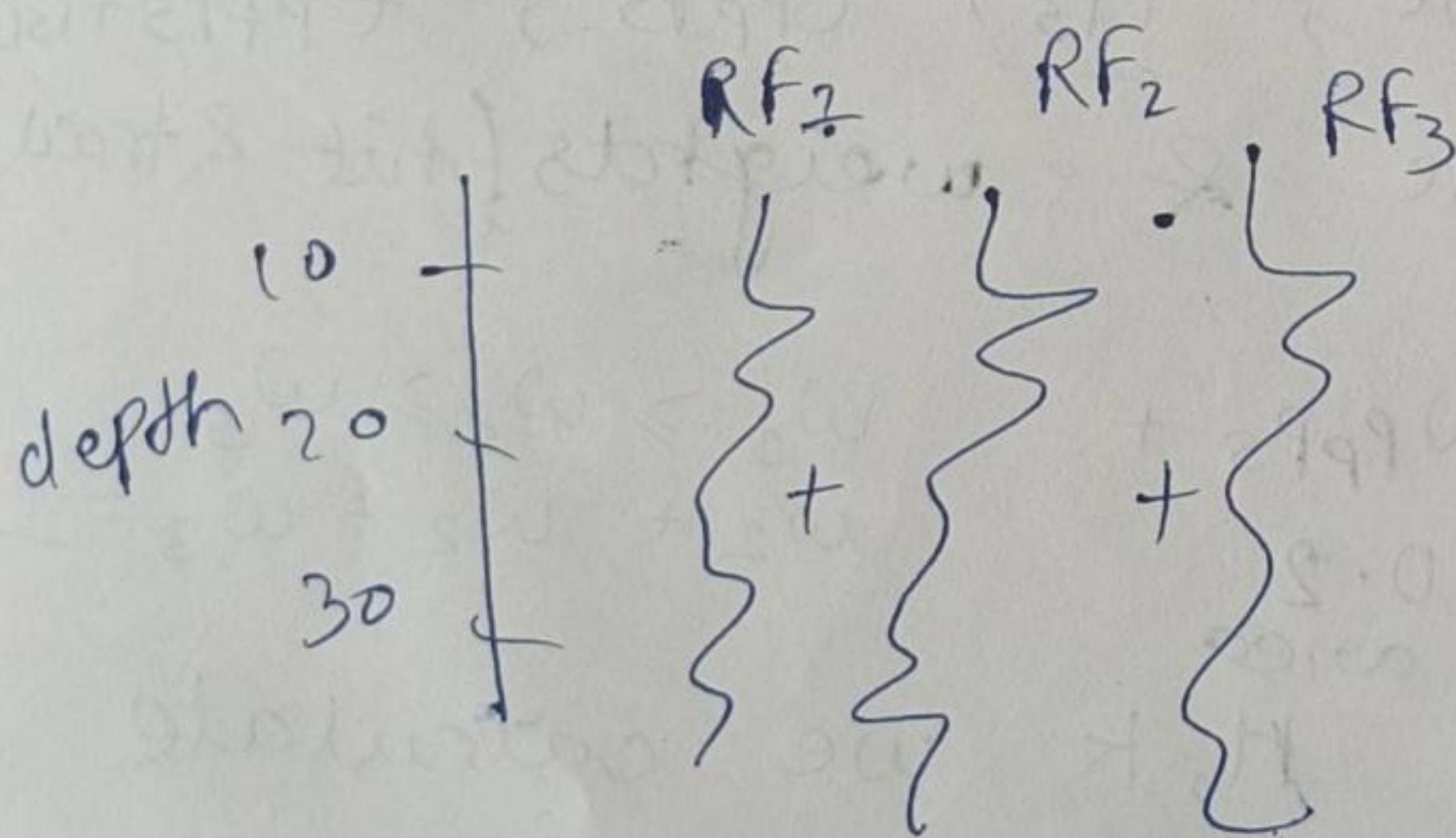
hence we obtain depth. 0.3

$\alpha_1 \beta_1$

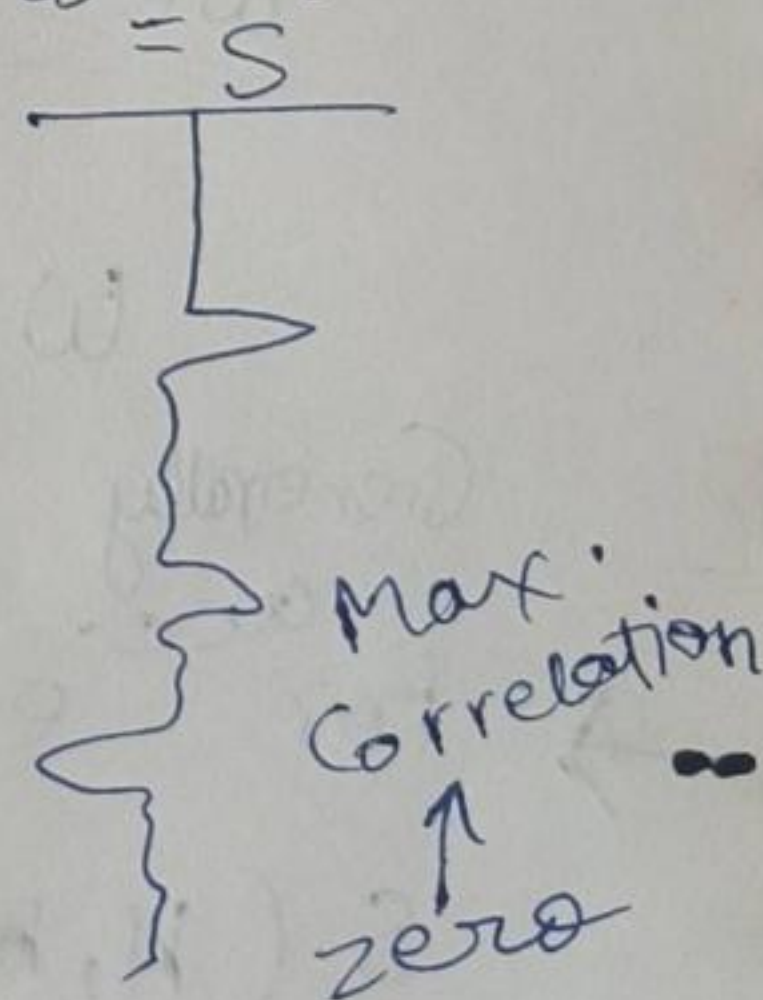
$\alpha_2 \beta_2$

$\alpha_3 \beta_3$

Assumed Velocity Model.



Master Trace = S



$$CC_j = \frac{\sum_{i=1}^N S * R_{fi}}{N}$$

$$= CC_1$$

Then summing them

Cross Correlation of Master trace with RF_1, RF_2 & RF_3

again we change the velocity model &
 rep again we get CC_2 , ~~another~~ CC_3
 and compare CC_1, CC_2

if $CC_2 > CC_1 \rightarrow$ Reject CC_2

$CC_3 < CC_1 \rightarrow$ Update to CC_3

$\rightarrow$ How to generate velocity Model:
 ↓
 * Optimisation Method

Here we get
 Moho depth

$\rightarrow$ The converged CC value will give the
 optimized depth.

$\rightarrow$ Lithosphere
 Asthenosphere
 Boundary
 (Depth)

S_p Receiver function

$\rightarrow$ Then CC_j will be Joint cross -
 correlation error

$$= \sum_{i=1}^{N_{Ps}} \frac{S_{Ps} * R_{F_{Ps}}}{N_{Ps}} + \sum_{j=1}^{N_{Sp}} \frac{S_{Sp} * R_{Sp}}{N_{Sp}}$$